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PALEOMAGNETISM OF 1780–1770 MA MAFIC AND COMPOSITE INTRUSIONS OF SMÅLAND (SWEDEN): IMPLICATIONS FOR THE MESOPROTEROZOIC SUPERCONTINENT

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ABSTRACT. We present a paleomagnetic study of 1785 to 1770 Ma composite and mafic intrusions in Småland, Sweden. A positive inverse baked contact test suggests the primary nature of their magnetization. The palaeomagnetic pole (45.7°N, 182.7°E, A_{95}=8.0°) may be regarded as a key pole for Fennoscandia.

Available paleomagnetic data suggest that Laurentia and Fennoscandia/Baltica drifted together since 1780 Ma until at least 1265 Ma. These two continents could form a core of a hypothetical Mesoproterozoic supercontinent.

Key words: Paleoproterozoic, Fennoscandia, Trans-scandinavian Igneous Belt, Sarmatia, paleomagnetism, Småland porphyries, Nuna supercontinent

INTRODUCTION

The hypothesis of Precambrian supercontinents has been strongly debated in recent decades (for example, Piper, 1976, 2000; Rogers, 1996; Condie, 2000; Meert, 2002; Pesonen and others, 2003; Zhao and others, 2004). The recent UNESCO IGCP 440 project "Rodinia Assembly and Breakup" (Bogdanova and others, 2008a) marked a significant breakthrough in our understanding of the youngest Precambrian supercontinent, Rodinia. However, its exact configuration and history is still controversial (for example, Dalziel, 1997; Pisarevsky and others, 2003; Li and others, 2008, 2009), owing mainly to the small number of Neoproterozoic high-quality, well-dated paleomagnetic poles available. The situation for Mesoproterozoic and older times is even worse. Evans and Pisarevsky (2008) estimated the number of high-quality pre-800 Ma paleomagnetic poles (paleopoles) at less than fifty. This is not enough to construct adequate Apparent Polar Wander Paths (APWPs)—the major tool for Phanerozoic reconstructions. Hence it is necessary to rely primarily on the "key pole" approach (Buchan and others, 2000), which causes additional problems such as polarity ambiguity and longitudinal uncertainty. Although those problems can be partly resolved by comparison of coeval paleopoles from different continents (Cawood and others, 2006; Evans and Pisarevsky 2008), there are very few coeval pairs of paleopoles available.

Traces of the widespread Late Paleoproterozoic orogenic events in various continents persuade some workers to suggest the existence of the Mesoproterozoic supercontinent called Nuna by Hoffman (1997), but also known as Columbia or Hudsonland (for example, Condie, 2000; Rogers and Santosh, 2002; Meert, 2002; Pesonen and others, 2003; Zhao and others, 2004). Meert (2002) analyzed some non-key paleopoles and proposed that this supercontinent was assembled after *ca.* 1770 Ma. Some recent paleomagnetic data, however, suggest that some building blocks of Nuna (Laurentia and Baltica) were fixed with respect to each other from *ca.* 1750 Ma until *ca.* 1270 Ma (Irving and others, 2004; Salminen and Pesonen, 2007). These results imply a long-lived connection between Laurentia and Baltica with north-eastern (present coordinates) Baltica facing south-eastern Greenland (for example, Patchett and Bylund, 1977; Stearn and Piper, 1984; Park, 1992; Starmer, 1996). The proposed

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connection is supported by geological similarities between Archean crustal blocks and Proterozoic orogenic belts in two continents (for example, Winchester, 1988; Gower and others, 1990; Gorbatschev and Bogdanova, 1993). However, additional paleomagnetic tests are required to verify this connection.

Post-Svecofennian (*ca.* 1800-1750 Ma) rocks of Fennoscandia have undergone extensive paleomagnetic study (see Pisarevsky and Sokolov, 2001 for an overview). Despite the consistency of the paleopoles, most of them are not supported by robust evidence for the primary remanence and only a few are precisely dated. Consequently, 1870 to 1750 Ma Fennoscandian poles were usually not cited as "reliable" (for example, Evans and Pisarevsky, 2008), or as "key poles" (Buchan and others, 2000) used in paleogeographic applications.

Nilsson and Wikman (1997) dated two composite intrusions in Småland, Sweden (figs. 1A and 1B) at 1780 ± 3 and 1776 +8/−7 Ma (U-Pb zircon) correspondingly. Nilsson (1992) demonstrated that mafic magmatism in Småland is probably genetically related (and so coeval) to these composite intrusions. Here we present results of a paleomagnetic study of these and some composite mafic-felsic and mafic intrusions in the Småland area.

GEOLOGY AND SAMPLING

The studied area (fig. 1) belongs to the 1.81 to 1.76 Ga Småland-Värmland belt—the southern and oldest part of the Trans-scandinavian Igneous Belt (TIB) formed mainly by reworking of the juvenile 2.10 to 1.87 Ga Svecofennian crust in a continental-arc setting (Högdahl and others, 2004). In the west, the TIB is bound by the Eastern Segment of the 1.14 to 0.95 Ga Sveconorwegian orogen along the Sveconorwegian Frontal Deformational Zone, or Protogine Zone (for example, Gorbatschev and Bogdanova, 1993, 2004).

Mafic and felsic plutons and dikes are widespread in the Småland-Värmland belt. In the eastern parts of the Småland province a special set of composite dikes has been found. These dikes are mainly composed of a marginal mafic and a central felsic part. To varying degree magma mixing features between the two parts can be observed. In the literature they are described under the general name of "Småland dyke porphyries". Some of these dike porphyries do not have mafic margins (Nilsson, 1992). Composite dikes are often regarded as a final product in a complex evolution with interplay between coeval felsic and mafic magmas (for example, Frost and Mahood, 1987). The specific structure of the Småland porphyries with the mafic material generally along the dike walls, has been interpreted with a special type of intrusion mechanism (for example, Carrigan and others, 1992; Eichelberger, 1995) in which the mafic magma due to its low viscosity, has facilitated the transport of the felsic magma in the same way as water is used as lubricant along the walls in an oil pipe line. These relationships suggest that the mafic components have the same age as the radiometrically dated felsic parts.

Radiometric dating (U-Pb, zircons) of the felsic component in the composite dikes has been reported by Nilsson and Wikman (1997). Samples from the Påskallavik and Alsterbro intrusions have been analyzed, resulting in 1780 ± 3 Ma for the former and 1776 +8/−7 Ma for the latter. These ages are somewhat, but not significantly younger than the ambient granitoids confirming the conclusions (for example, Persson, 1985) that the dike porphyries are the final expression of the granite magmatism in the area.

A major, mainly geochemical, investigation of these and other related dikes has been made be Nilsson (1992). One of his results is that the mafic parts in the composite dikes have a similar chemistry as some mafic dikes without a felsic component, but with the same NW-trending direction in the northwestern part and mainly ENE-trending direction for the rest of the investigated area. This dike swarm, called the "Oskarshamn

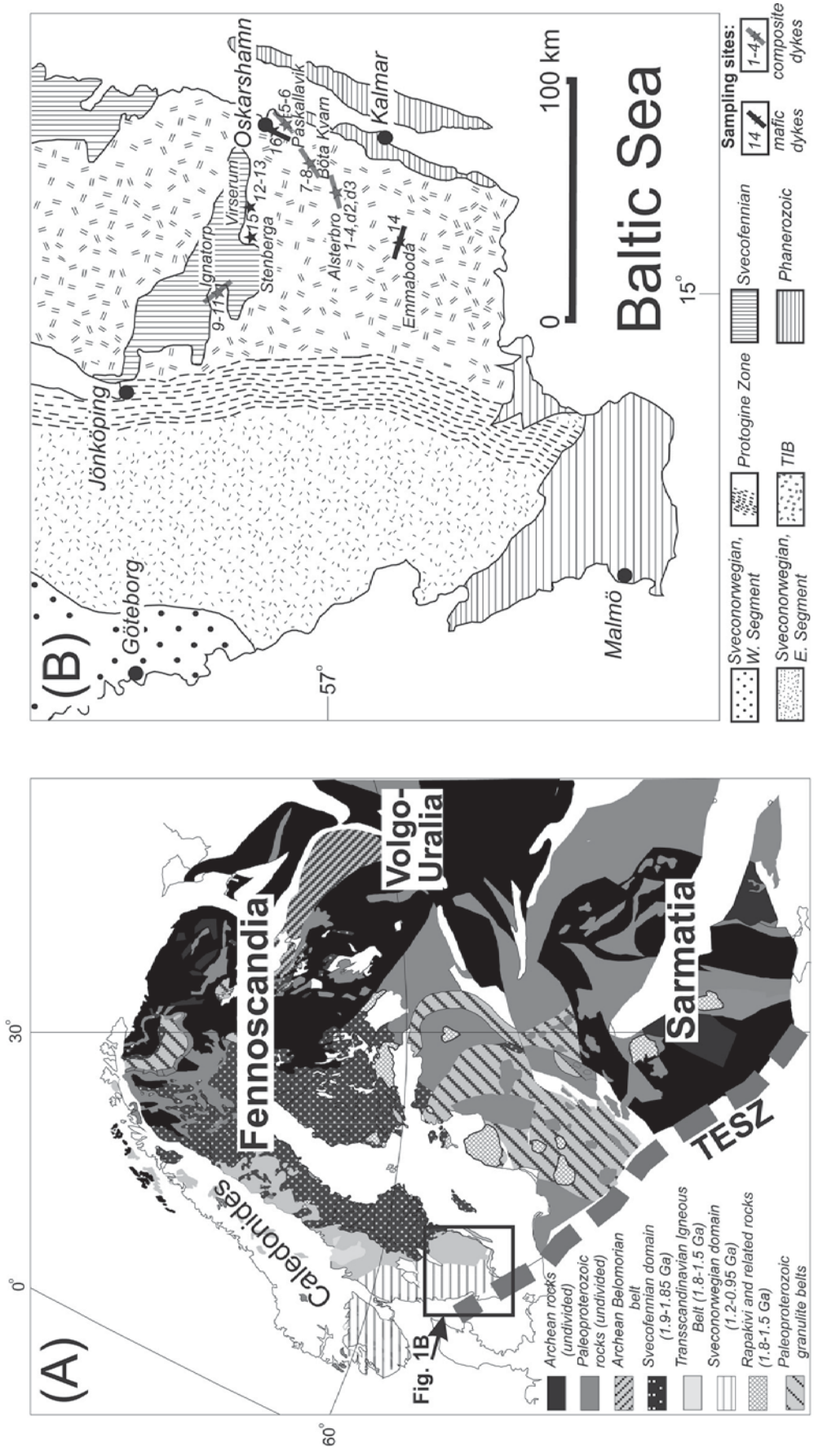


Fig. 1. (A) Geological map of Baltica (simplified from Bogdanova and others, 2001), TESZ—Trans-European Suture Zone; (B) geological sketch of southern Sweden and sampling localities.

dykes", is therefore inferred to be the same age as the composite dikes. A younger dike swarm, the "Vimmerby dykes" with a general NE-trending direction is geochemically different. Field observations confirming this relative age difference has also been made by Persson (1985), however the absolute age of the Vimmerby dykes is unclear. A few larger mafic intrusions also outcrop in the area. Our paleomagnetic data suggest that they are likely to be related to the Oskarshamn dykes.

Some mafic dikes of the 978 to 946 Ma Blekinge-Dalarna swarm (Söderlund and others, 2005) were found in the eastern part of the area and studied paleomagnetically (Bylund, 1985, 1992; Pisarevsky and Bylund, 1998).

In Alsterbro and Påskallavik (fig. 1B) we collected oriented block samples for the paleomagnetic study of both the composite intrusions dated by Nilsson and Wikman (1997). Additionally we sampled two more composite dikes (Ignatorp and Böta Kvarn), four mafic dikes (Emmaboda, Oskarhamn, and Alsterbro), and two large gabbroic bodies (Stenberga and Virserum). Sample locations are shown in figure 1B. Studied composite dikes are *ca.* 10 to 70 m thick, mafic dikes are 0.5 to 3 m thick. Nilsson (1992) reported that the felsic cores of the composite dikes are more resistant to alteration than the mafic edges, so we concentrated our efforts on the fine-grained groundmass of these felsic cores. A total of 150 oriented blocks were collected from 20 sites. Both magnetic and sun compasses were used for the orientation. Sampled localities are shown in figure 1B, and the coordinates are in table 1.

ANALYTICAL TECHNIQUES

In the laboratory from 2 to 12 cylindrical specimens (25 mm diameter) were drilled from each oriented block. During sampling and drilling of porphyries we tried to avoid large phenocrysts, so all our specimens are from the relatively fine-grained matrix.

Samples were analyzed in the paleomagnetic laboratory of Lund University (Sweden) and in the Tectonics Special Research Centre at the University of Western Australia. Magnetic remanence was measured using 2G cryogenic magnetometers and a JR-5 spinner magnetometer (Geofyzika, Brno, Czech Republic.) Stepwise thermal demagnetizations, up to 600 to 650°C, and progressive alternating field (AF) demagnetizations, up to 160 mT, were applied using Schonstedt TSD-1, Magnetic Measurements TD2 furnaces, and a 2G600 automated degaussing system. Normally 3 or 4 cylinders (in some cases up to eight) from each oriented block were demagnetized. The mean characteristic remanence (ChRM) directions for each oriented block were calculated and used for the site mean calculations (table 1). Magnetic mineralogy was investigated from demagnetization characteristics, saturation isothermal remanent magnetization (SIRM) curves and, in selected samples, from detailed variation of susceptibility versus temperature (20 to 700°C) obtained using the Bartington meter in conjunction with an automated Bartington furnace. SIRM characteristics were studied by inducing fields with a Molspin Ltd magnetic pulse charger for fields up to 200 mT and a Redcliffe 700 BSM pulse charger for fields up to 4 T.

PALEOMAGNETIC ANALYSES

Composite Intrusions (Alsterbro, Ignatorp, Böta Kvarn, Påskallavik)

The intensity of the natural remanent magnetization (NRM) of porphyries ranges from 2 to 200 mA/m, and their magnetic susceptibility from about 1 to 20×10^{-3} SI units. Susceptibility versus temperature study (fig. 2) demonstrates that magnetite is the main magnetic carrier in these rocks. Small Hopkinson peaks indicate that some proportion of magnetite is single domain (SD). SIRM curves (fig. 3A) indicate a strong coercivity of porphyries probably caused by SD magnetite and hematite. Thermal and AF demagnetizations of NRM confirm this suggestion and demonstrate that magnetite

Table 1

Paleomagnetic directions and poles, TIB1 rocks, Småland, Sweden

#	Site	Locality	Rocktype	N/n	Slat (°N)	Slong (°E)	Decl. (°)	Incl. (°)	k (°)	α_{95} (°)	Plat (°N)	Plong (°E)	D_p (°)	D_m (°)
1	1	Alsterbro	porphyry	6/7	56.93	15.92	27.1	28.8	16.6	16.9	43.7	158.5	10.2	18.6
2	2	Alsterbro	porphyry	4/6	56.93	15.92	12.3	36.5	39.8	14.9	52.2	176.4	10.1	17.4
3	3	Alsterbro	porphyry	4/5	56.93	15.92	17.9	35.1	12.4	27.2	50.1	169.0	18.1	31.4
4	4	Alsterbro	porphyry	5/5	56.93	15.92	31.0	20.6	41.8	12.0	37.9	156.0	6.6	12.6
5	7	Bota Kvärn	porphyry	3/3	57.03	16.13	0.9	23.4	18.6	29.4	45.2	194.9	16.8	30.6
6	8	Bota Kvärn	porphyry	3/3	57.03	16.13	4.1	24.6	57.5	16.4	45.8	190.4	9.4	17.6
7	9	Ignatorp	porphyry	5/6	57.60	15.40	13.3	18.1	15.2	20.3	40.6	178.0	10.9	21.1
8	10	Ignatorp	porphyry	5/5	57.60	15.40	9.5	23.5	14.8	20.6	44.1	182.4	11.7	22.0
9	11	Ignatorp	porphyry	7/7	57.60	15.40	7.6	7.8	22.0	13.2	36.0	186.0	6.7	13.3
10	14	Emmaboda	dolerite	10/13	56.63	15.55	352.1	33.5	49.1	7.0	51.2	207.6	4.5	8.0
11	15	Stenberga	gabbro	6/16	57.30	15.50	166.8	-18.2	44.2	10.2	41.0	212.9	5.5	10.6
	Mean pole for 11 sites										***45.7***	***182.7***	***A_{95}=8.0***	
	Mean pole for 9 sites (without sites 3 and 7)										***45.1***	***182.7***	***A_{95}=9.6***	
12	16	Oskarhamn	dolerite	8/18	57.30	16.20	195.9	45.8	122.0	5.0	4.4	182.2	4.1	6.4
13	d2	Alsterbro	dolerite	7/7	56.93	15.92	210.6	48.7	8.8	21.6	-0.3	169.6	18.7	28.4
14	d2	Alsterbro	b. contact	3/3	56.93	15.92	207.7	28.2	4.8	63.7	14.5	168.3	38.3	69.8
15	d3	Alsterbro	dolerite	6/6	56.93	15.92	297.0	-48.1	41.8	10.5	-11.0	248.4	9.0	13.7
16	d3	Alsterbro	b.contact	2/2	56.93	15.92	211.3	-66.1	-	-	-69.5	296.8	-	-

N/n=number of independently oriented blocks used/measured; Slat, Slong=locality latitude and longitude; Decl, Incl=site mean declination, inclination; k=best estimate of the precision parameter of Fisher (1953); a_{95}=the semi-angle of the 95% cone of confidence; Plat, Plong=latitude, longitude of the paleopole; Dp, Dm=the semi-axes of the cone of confidence about the pole at the 95% probability level.

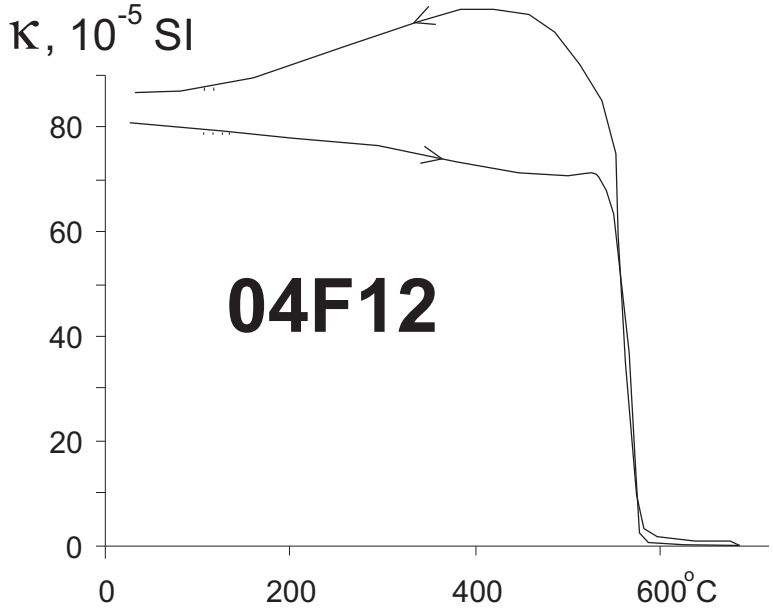


Fig. 2. Analysis of Curie points using magnetic susceptibility versus temperature curves (porphyry, Ignatorp).

and hematite remanence components have very similar directions (figs. 4A and 4B). After removal of a low-stability viscous overprint, both thermal and AF demagnetization isolated a single stable ChRM in three intrusions (Alsterbro, Ignatorp, Böta Kvarn). This ChRM has shallow downward (very shallow upward in a few samples) N-NNE direction (figs. 4A and 4B; table 1, entries 1-9). Demagnetization behavior of porphyries from the Påskallavik intrusion is similar to three other intrusions, but the directions are very scattered, so we excluded this body from overall means calculations and we do not show these data in table 1.

Dolerite Dike (Emmaboda)

The intensity of NRM of the dolerites ranges from 10 to 1000 mA/m, and their magnetic susceptibility is around 0.5 to 1.0×10^{-3} SI units. SIRM curves (fig. 3B) demonstrate that the saturation fields are >400 mT, which is characteristic for the SD-PSD magnetite. This inference is supported by "square shoulder" thermal demagnetization curves (fig. 4C). Apart from viscous remanence, only one magnetite-bearing NRM component with N-NNW medium to shallow downward direction was isolated (fig. 4C; table 1, entry 10). All studied samples yield coherent ChRM.

Gabbroic Intrusions (Stenberga, Virserum)

The intensity of NRM of the Stenberga gabbro varies from 100 to 9000 mA/m, their magnetic susceptibility is between 0.7 and 1.4×10^{-3} SI units. These rocks are magnetically softer than dolerites (fig. 3C) and their NRM is mostly carried by the multi-domain (MD) magnetite. Two remanence components were isolated during thermal demagnetization. The first component is scattered around the present day field direction (PDF) and only this component was isolated in 7 strongly (NRM > 3 A/m) magnetized samples. However, the second SSE shallow upward component was also isolated in 5 weaker magnetized samples (fig. 4D). We consider this component as the ChRM. Its mean direction is nearly opposite to the ChRM directions of other studied intrusions (fig. 5; table 1, entry 11). The result of the reversal test of McFadden and McElhinny (1990) is indeterminate.

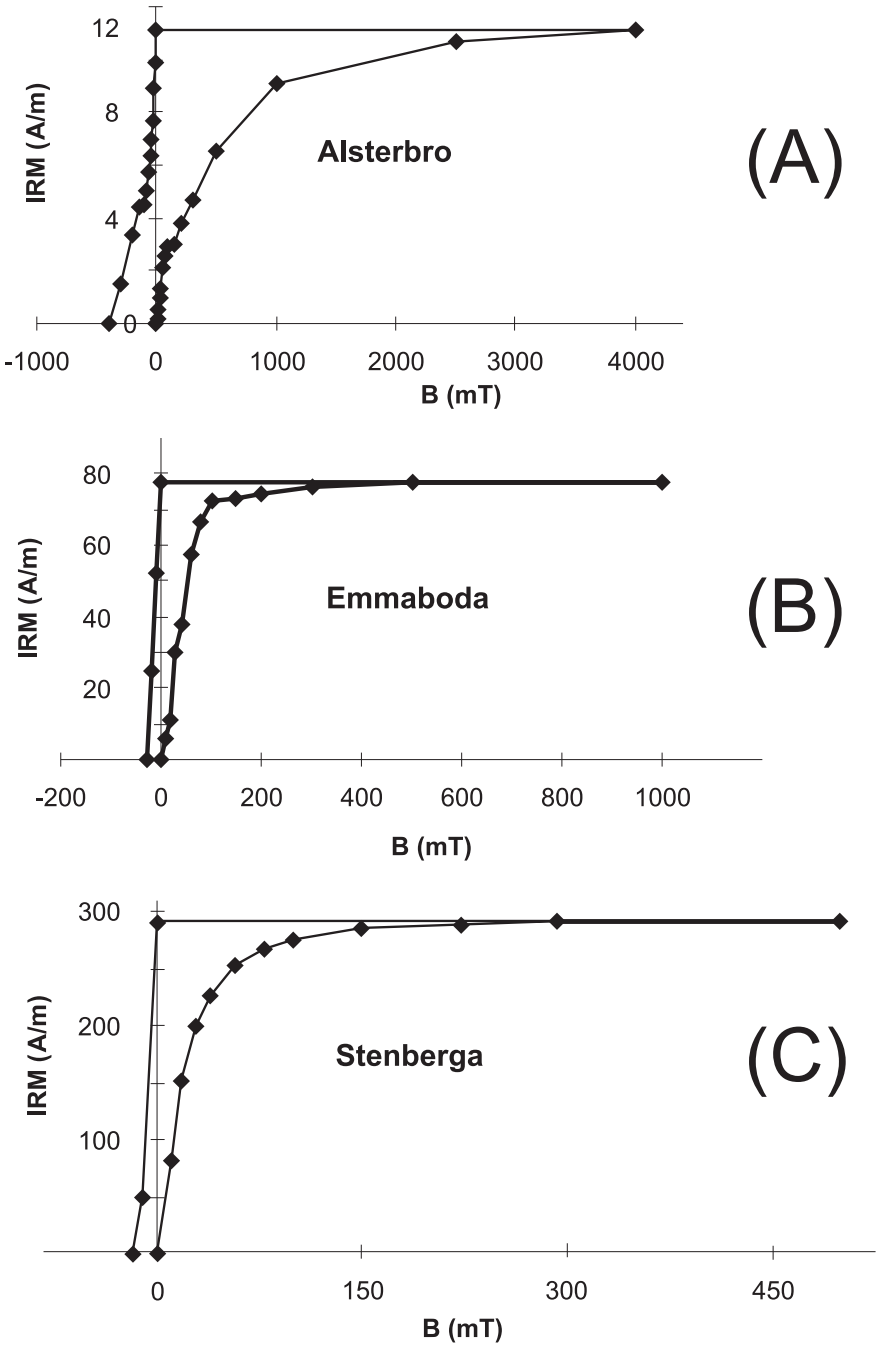


Fig. 3. IRM saturation curves: (A) porphyry; (B) dolerite; (C) gabbro.

The intensity of NRM of the Virserum gabbro is much weaker and no stable remanence was isolated.

Dolerite Dike, Oskarhamn

The intensity of NRM ranges from 70 to 1200 mA/m. This dike carries a stable single-component ChRM with moderately downward SSW direction (fig. 6, table 1, entry 12), which is significantly different from all above-mentioned remanence directions. We suggest that although this dike outcrops near Oskarhamn, it does not belong to the "Oskarhamn" swarm, but is significantly younger (see DISCUSSION).

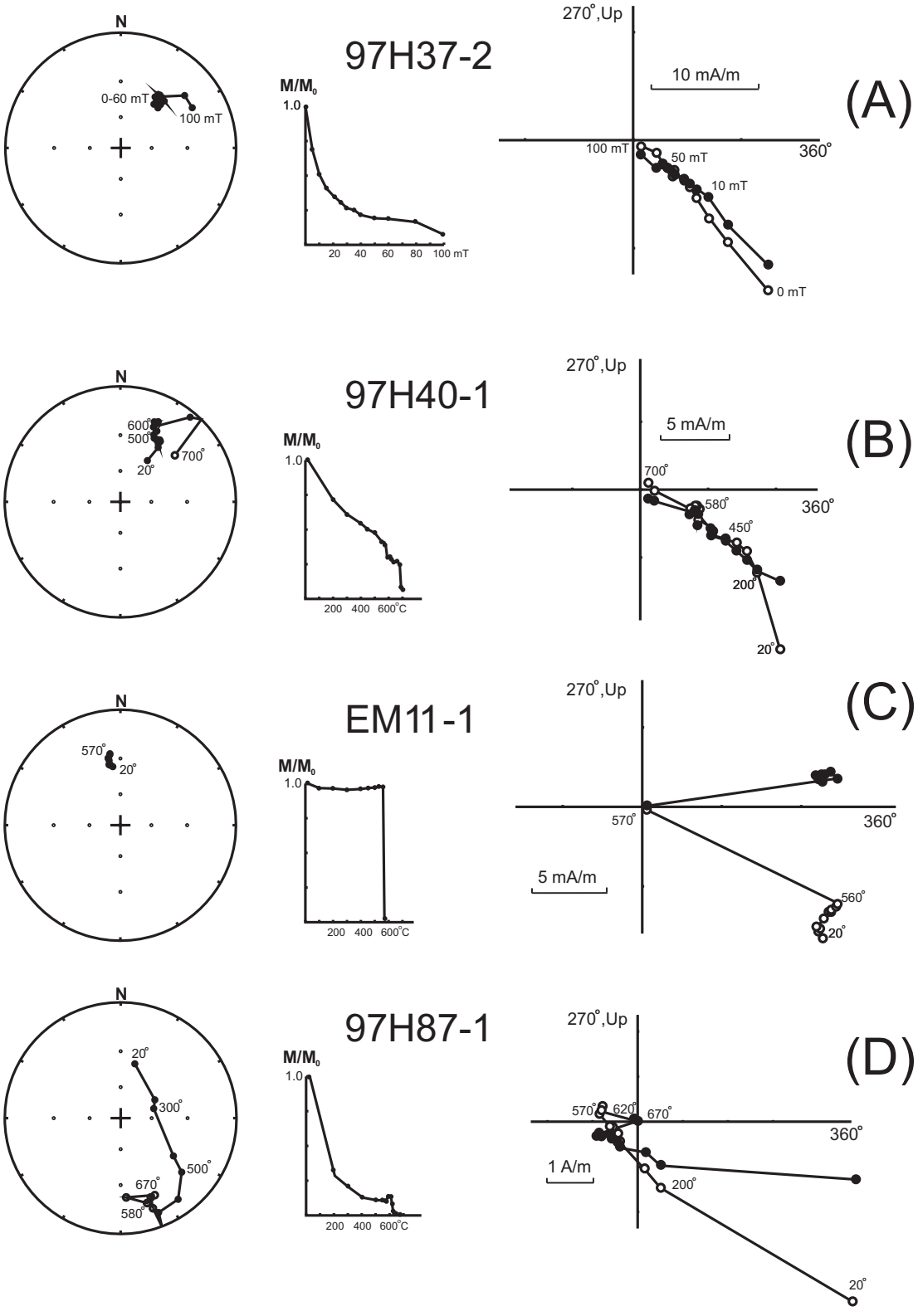


Fig. 4. Stereoplots and examples of demagnetization behavior for studied rocks. In orthogonal plots, open (closed) symbols show magnetization vector endpoints in the vertical (horizontal) plane; curves show changes in intensity during demagnetization. Stereoplots show upwards (downwards) pointing paleomagnetic directions with open (closed) symbols. (A) porphyry, AF—demagnetization; (B) porphyry, thermal demagnetization; (C) Emmaboda dolerite; (D) Stenberga gabbro, stronger magnetization.

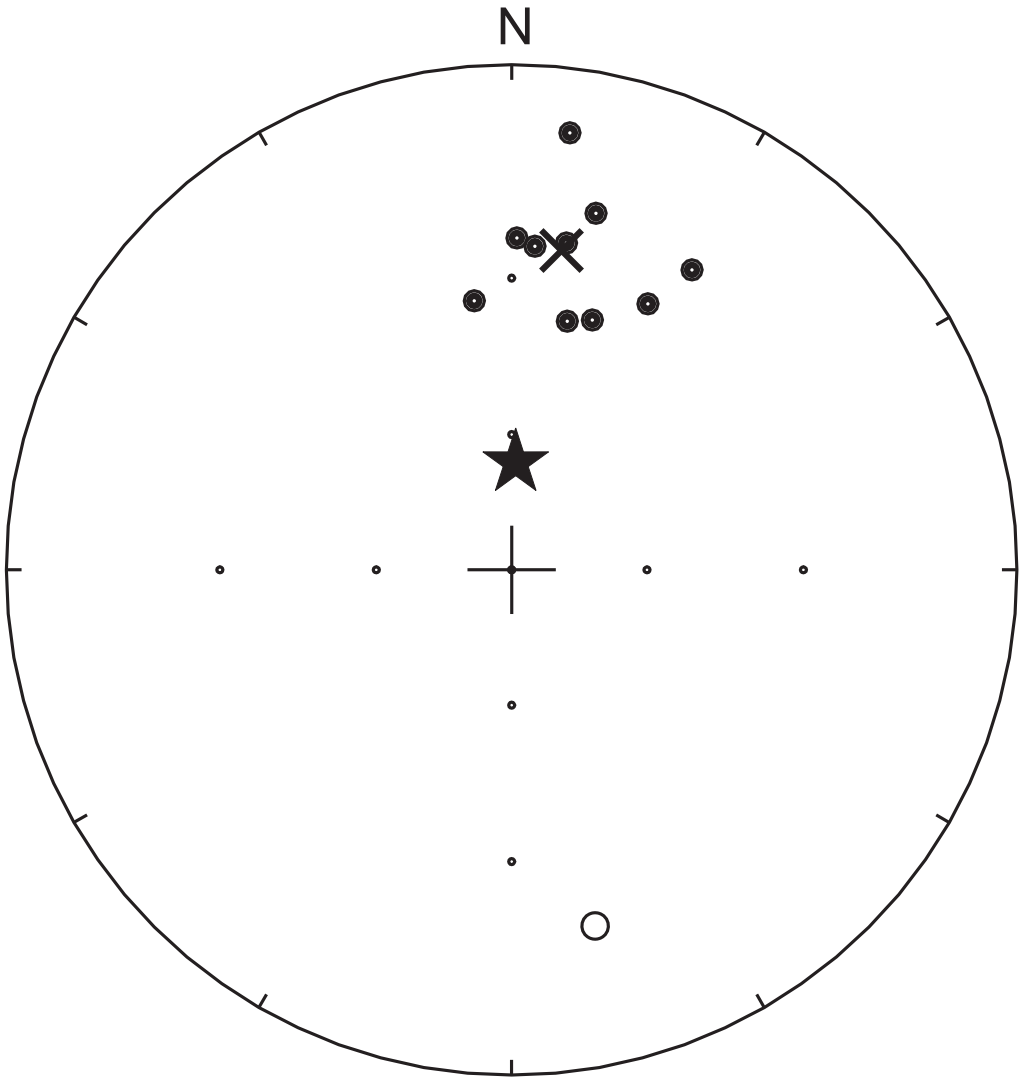


Fig. 5. Stereoplot of the site mean directions of ChRM, interpreted to be primary in this study; sites 1–4, 7–11, 14, 15; star=present day geomagnetic field direction, cross=mean ChRM direction for all sites.

Alsterbro Minor Dikes, Inverse Contact Test

Thermal and AF demagnetizations of the sampled country granites did not isolate any stable ChRM, so the direct contact test was impossible.

However, stable ChRMs were found in two thin mafic dikes intruded into the Alsterbro composite intrusion. In addition, exocontacts of these thin dikes were completely remagnetized during their emplacement. The magnetic properties of these dikes are similar to other dolerites in this study and one ChRM component was isolated in each dike. The ChRM directions differ from each other and from the mean direction of the main intrusion (figs. 7A, 7C, and 7E; table 1, entries 13, 15). Dike 3 (fig. 7A) is apparently "reversely" (upward) magnetized and its mean remanence direction resembles the "Sveconorwegian" direction of the 978 to 946 Ma Blekinge–Dalarna dikes (for example Bylund, 1985, 1992; Pisarevsky and Bylund, 1998; Bylund and Pisarevsky, 2002). In particular, the remanence direction of dike 3 (D=297°; I=−48°) is very close to the remanence direction of nearby Åseda dolerites and they have the same polarity (D=304°; I=−38°; Pisarevsky and Bylund, 1998). The remanence of dike 2 is more scattered, has an opposite polarity with respect to the ChRM of

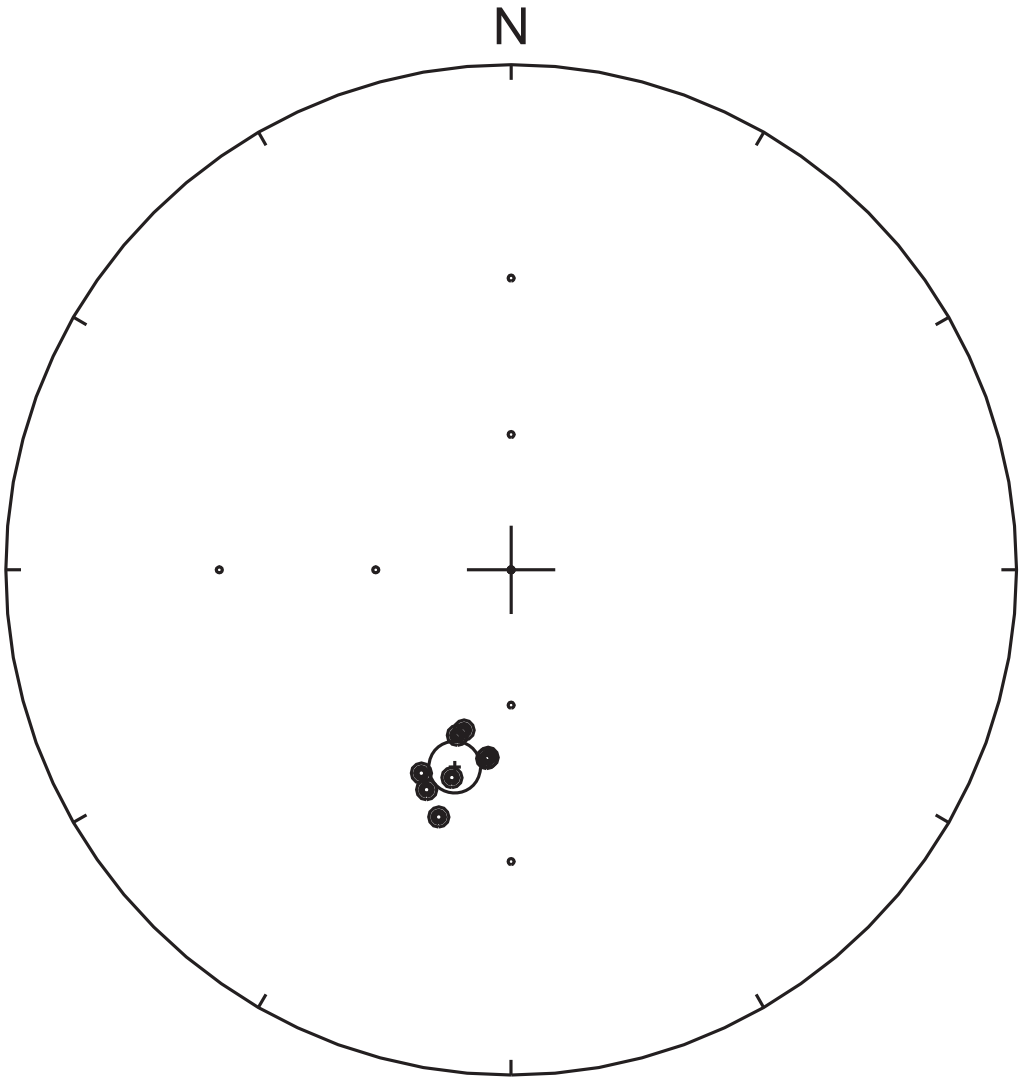


Fig. 6. Stereoplot of the sample mean directions of ChRM in the Oskarhamn dolerite dike (site 16).

dike 3, but not antiparallel to it (fig. 7). This polarity change implies that the time difference between emplacements of the two dikes was sufficiently large to accommodate at least one geomagnetic reversal in between. Notably, the ChRM direction in dike 2 is similar to that in the above mentioned dolerite dike near Oskarhamn (table 1, entry 12). Baked porphyries are apparently re-magnetized during emplacement of the dikes (figs. 7B and 7D; table 1, entries 14, 16). The difference between ChRM direction of dikes and porphyries suggests that both studied dikes are significantly younger than the composite intrusion and that the ChRM of porphyries predates the intrusion of smaller dikes (positive inverse baked contact test).

DISCUSSION

ChRM directions of three out of four studied composite intrusions are coherent and supported by the baked contact test. This direction corresponds to the time of their emplacement at 1785 to 1770 Ma. The Påskallavik composite intrusion revealed a highly scattered ChRM and was excluded from the further analyses. Both thermal and AF demagnetizations of dolerite and gabbro (sites 14 and 15) isolated stable ChRMs with opposite polarities and with similar (but not identical) directions to those of the

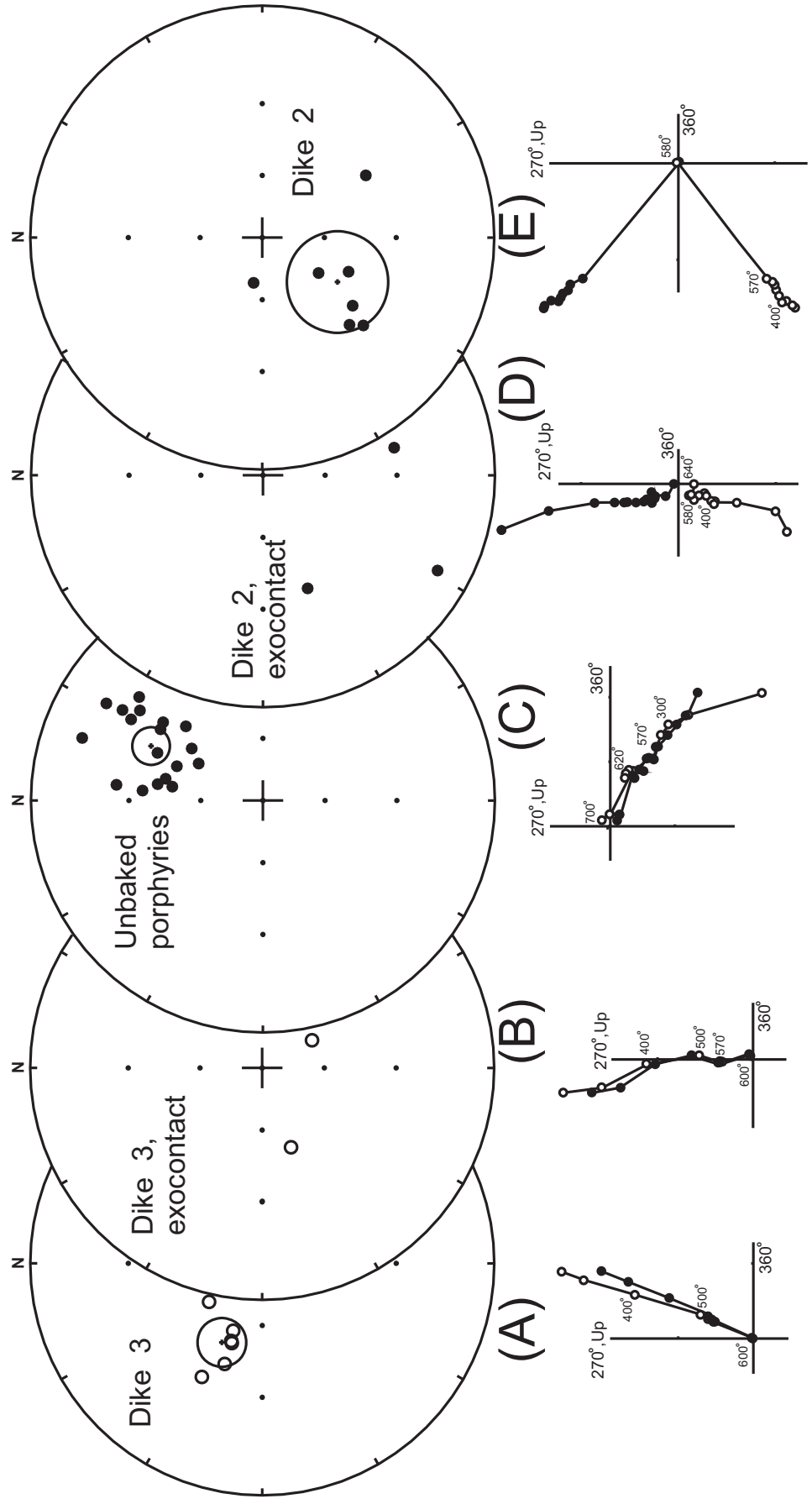


Fig. 7. Baked contact test in the Alsterbro composite intrusion; stereoplots and examples of orthogonal plots: (A) dolerite dike 3; (B) baked porphyries at the contact of dike 3; (C) porphyries distal from dolerite dikes; (D) baked porphyries at the contact of dike 2; (E) dolerite dike 2.

composite intrusions. This similarity suggests that the Emmaboda dike belongs to the Oskarhamn swarm (Nilsson, 1992) and confirms the above mentioned suggestion that mafic dikes of this swarm and Småland gabbroic intrusions are only slightly younger than the composite intrusions. Taking this into account we calculated the mean paleopole for the studied Småland composite and mafic intrusions (table 1), which we consider to provide the *ca.*1780 Ma key pole for Fennoscandia.

On the other hand, the dolerite dike near Oskarhamn and minor dike 2 in Alsterbro, whose primary remanence is supported by the baked contact test, are significantly younger. These dikes are not dated, but the calculated paleopoles (table 1, entries 12 and 13) are close to the mean 1460 Ma paleopole for Baltica (table 2, entry 4). However, these paleopoles should be treated cautiously because of their rather uncertain age constraints. The primary (supported by the contact test) ChRM of minor dike 3 in Alsterbro is similar to the remanence directions of the 978 to 946 Ma Blekinge-Dalarna mafic dikes (Söderlund and others, 2005), and in particular to the remanence direction of the Åseda dike (Pisarevsky and Bylund, 1998) located just ~30 km NW of Alsterbro. Hence we tentatively assume the 978 to 946 Ma age for the Alsterbro dolerite dike 3.

Buchan and others (2000) did not find a single key pole from Fennoscandia older than ~1640 Ma. Pisarevsky and Sokolov (2001) reported a study of the Shoksha Formation redbeds supported by a conglomerate test and by the presence of the magnetostratigraphy pattern (SH pole in table 2, entry 2). Pesonen and others (2003) considered this pole as a key pole. The Shoksha Formation predates the 1770 ± 12 Ma Ropruchey sill (Bibikova and others, 1990) and postdates the ~1800 Ma underlying Petrozavodsk Formation (Sokolov and others, 1987). However the latter date (K-Ar on glauconite) is not considered as very reliable, so the exact age of this pole is unclear. Our new *ca.*1780 Ma pole from the Småland intrusions has better age constraints and is supported by the inverse contact test (SM pole in table 2, entry 1). The angular difference between SH and SM poles is 28° and probably reflects their age difference.

The similarity to the combined paleopole of the 1770 Ma Ropruchey Sill in Karelia, Russia (Damm and others, 1997; Fedotova and others, 1999) (RP pole in table 2, entry 3) suggests that the age of the SH pole is closer to the younger limit of the original 1790 to 1770 Ma estimation by Pisarevsky and Sokolov (2001).

Reliable Late Paleoproterozoic and Mesoproterozoic paleopoles are scanty and not sufficient for unequivocal paleogeographic reconstructions. Table 2 clearly demonstrates this for Laurentia and Baltica, and the situation for most of the other continents is worse (for example, Evans and Pisarevsky, 2008). However, these poles can be used for testing some reconstructions based on geological data. We shall follow this approach here.

Salminen and Pesonen (2007) obtained a new robust 1460 Ma paleopole for Baltica and proposed a new reconstruction of Laurentia and Baltica in the Mesoproterozoic between 1460 and 1270 Ma. Moreover, following the suggestion of Pesonen and others (2003), Salminen and Pesonen (2007) used new paleomagnetic data from 1740 Ma Cleaver Dykes (Irving and others, 2004) and the above mentioned SH Baltican pole to propose that Laurentia and Baltica could have been connected by 1790 to 1770 Ma. Interestingly, the Laurentia-Baltica reconstruction of Salminen and Pesonen (2007) is similar, but not identical to the reconstruction of Gower and others (1990) built by matching pre-Neoproterozoic crustal blocks and orogenic belts of these cratons. The difference between the "geological" reconstruction of Gower and others (1990) and "paleomagnetic" reconstruction of Salminen and Pesonen (2007) is minimal and we think that they are within the precision limits of paleomagnetic and geochronological methods.

TABLE 2

Selected paleomagnetic poles from Baltica (Fennoscandia) and Laurentia

#	Abbr. name in Fig. 8	Rockname	Age (Ma)	Plat (°N)	Plong (°E)	A95 (°)	Reference	GPMDB #
				Fennoscandia, Baltica				
1	SM	Småland intrusions	1784-1769	45.7	182.7	8.0	This study	
2	SH	Shoksha sediments	1790-1770	39.7	221.1	4.0	Pisarevsky and Sokolov (2001)	8681
3	RP	Ropruchey sill	1782-1758	39.1	216.6	6.7	Fedotova and others (1999), Damm and others (1997)	7400, 7406-7408
4	BA1460	Mean for Baltica	~1460	15.0	181.0	11.0	Salminen and Pesonen (2007)	
5	BA1265	Mean for Baltica	~1265	4.0	158.0	4.0	Pesonen and others (2003)	
				Laurentia				
6	LA1750	Mean for Laurentia	1760-1740	20.8	265.5	5.2	Irving et al. (2004)	
		St. Francois Mts.	1476±16	-13.2	219.0	6.1	Meert and Stuckey (2002)	8932
		Michikamau Intr.	1460±5	-1.5	217.5	4.7	Emslie and others (1976)	2274
		Harp Lake Complex	1450±5	1.6	206.3	4.3	Irving and others (1977)	2055
7	LA1460	Mean for Laurentia	~1460	-4.4	214.2	16.0	This study	
8	MK	MacKenzie dykes	1267+7/-3	4.0	190.0	5.0	Buchan and Halls (1990), LeCheminant and Heaman (1989)	
				Sarmatia				
9	KO	Korosten pluton	1770-1740	25.9	170.7	3.8	Elming and others (2001)	8660-8668

GPMDB #: Result number in the Global Paleomagnetic Database (McElhinny and Lock, 1996), version 4.6 (Pisarevsky, 2005).

Unfortunately Gower and others (1990) did not provide Euler rotation parameters for their reconstruction. We tried to reproduce their reconstruction to the best of our abilities and deduced the following Euler rotation parameters: Baltica was rotated with respect to Laurentia at 45° anticlockwise around the Euler pole of 45°N, 7.5°E. We applied this Euler pole for the paleogeographic reconstructions shown in figure 8 (table 3). Clearly these reconstructions fit paleomagnetic data (table 2) reasonably well. There are some minor differences between our selection of paleopoles (table 2) and the choice of Salminen and Pesonen (2007). First, instead of the 1740 Ma Cleaver Dykes pole (Irving and others, 2004), we used the mean 1760 to 1740 Ma Laurentian pole calculated by the same authors. Second, we excluded the Spokane Formation pole (Vitorello and Van der Voo, 1977) from the 1460 Ma mean Laurentian pole. This pole was subsequently considered to represent a burial remagnetization (Elston and others, 2002).

A comparison of the 1770 to 1740 Ma paleomagnetic data from Ukraine (Elming and others, 2001) with non-key Fennoscandian poles suggests that Baltica was not yet assembled at that time. This interpretation places some constraints on the timing of the collision between Sarmatia-Volgouralia and Fennoscandia (Bogdanova, 1993; Gorbatschev and Bogdanova, 1993; Bogdanova and others, 2008b) and we included our interpretation of this event in the reconstructions (fig. 8). A similar model has been recently published by Elming and others (2010).

Table 2 shows that there are two roughly coeval pairs of Laurentian and Baltican paleopoles at 1460 and 1265 Ma, but the situation is more complicated for the 1780 to 1750 Ma time interval. Age mismatches between Laurentian and Fennoscandian poles exceed ten million years, so we cannot use their exact matching for the reconstructions. Instead we made a general assumption that Laurentia and Fennoscandia were parts of a single continent and their mutual configuration did not change for the whole time interval considered. The above mentioned estimated Euler rotation parameters were used to reconstruct this configuration. Then we reconstructed this joint continent (which could be part of some larger continent) paleomagnetically for three time slices using 1780 Ma (SM pole), 1770 Ma (RP and SH poles), and 1750 Ma (LA1750 pole). These three reconstructions show reasonably smooth drift of Laurentia-Fennoscandia and demonstrate that their fit is paleomagnetically permissive. A simplified (after Li and others, 2008) geological legend is also shown in those reconstructions. The chain of 1.8 to 1.6 Ga orogens (shown in black) suggests that the joint continent had a long-lived western active margin, as was proposed previously (for example, Karlstrom and others, 2001).

The Sarmatian KO paleopole is shown in the 1750 Ma reconstruction and demonstrates that the final collision between Fennoscandia and Sarmatia-Volgouralia was not completed at that time. The final docking probably occurred at *ca.* 1700 Ma (Bogdanova and others, 2008b; Elming and others, 2010), forming Baltica.

Paleomagnetic reconstructions at 1460 and 1265 Ma (fig. 8) demonstrate that the proposed Laurentia-Baltica configuration may have persisted for about 500 m.y. We suggest that these two continents could form a core of a hypothetical Mesoproterozoic supercontinent Nuna (Columbia, Hudsonland). Wingate and others (2009) reported that Siberia also could have been part of this supercontinent by at least *ca.* 1500 Ma according to new high-quality paleomagnetic data. Importantly, their paleomagnetically permissive *ca.* 1475 Ma reconstruction (see fig. 9 of Wingate and others, 2009) shows Siberia not in close connection with Laurentia-Baltica. That reconstruction leaves a space for another continent (or continents) of the size of Laurentia between Siberia and Laurentia-Baltica, making the hypothetical Nuna comparable to Gondwana. Therefore the recent paleomagnetic data provide a strong support for the Nuna

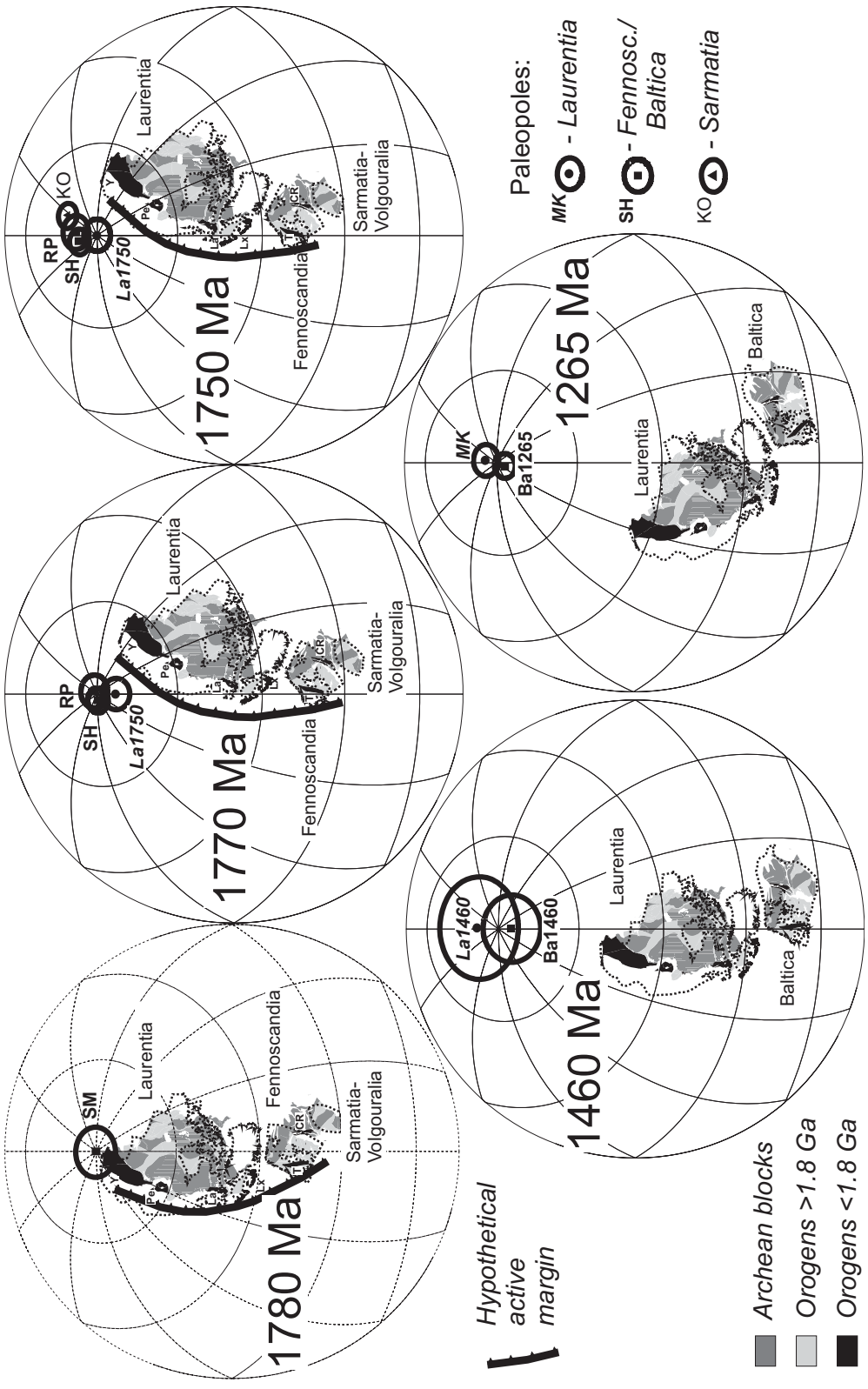


Fig. 8. Reconstructions of Laurentia, Fennoscandia, and Sarmatia at 1780–1265 Ma. The geological legend is simplified from the IGCP 440 project (Li and others, 2008). 1.8–1.6 Ga accretionary/collisional events: Y—Yavapai province, Pe—Penokean, La—Labrador, Lx—Laxfordian, CR—Central Russian Belt, T—TIB (Karlstrom and others, 2001; Trewin and Rollin, 2002; Bogdanova and others, 2008).

TABLE 3

Euler rotation parameters for 1780–1265 Ma reconstructions in Fig. 8

Craton	Age (Ma)	Pole Lat(°)	Pole Lon(°)	Angle (°)
Laurentia to absolute framework	1780	2.92	-32.31	-57.32
	1770	10.77	-17.29	-76.77
	1750	14.27	-14.48	-71.78
	1460	8.65	136.16	88.08
	1265	21.42	122.39	89.97
Fennoscandia/Baltica to Laurentia	1780-1265	44.99	7.45	44.93
Sarmatia to Fennoscandia	1780	46.92	49.92	26.11
	1770	52.07	41.63	30.76
	1750	52.43	39.28	34.94
Greenland to Laurentia	1780-1265	70.40	265.90	-18.00
Rockall to Laurentia	1780-1265	75.31	159.59	-23.46
Scotland terranes to Laurentia:				
Grampian	1780-1265	79.85	193.48	-34.77
Outer Hebridian-Torridonian	1780-1265	79.56	357.46	-47.00
Moine	1780-1265	81.15	353.53	-44.61
Sgurr Beag	1780-1265	86.16	318.40	-39.89

hypothesis. However, many more Mesoproterozoic paleopoles are required for further testing and verification of this hypothesis.

CONCLUSIONS

— A paleomagnetic study of Småland composite and mafic intrusions resulted in a new robust 1780 Ma palepole for Fennoscandia.
— This study, in conjunction with other paleopoles from Baltica and Laurentia, suggest that their mutual position and orientation was unchanged since 1780 Ma until 1265 Ma.
— Baltica and Laurentia could form a core of the hypothetical Mesoproterozoic supercontinent Nuna.

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